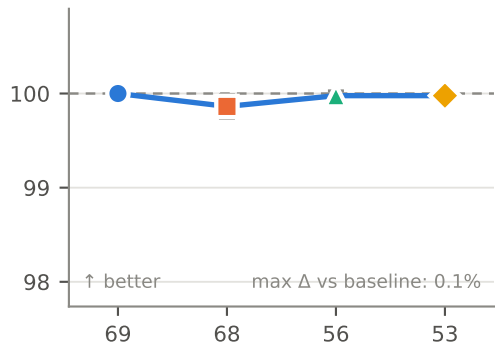


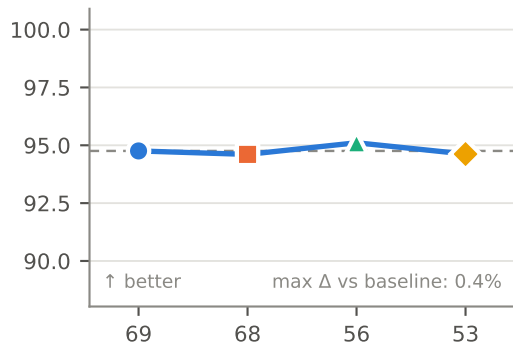
Observation ablation causes no measurable degradation

Isaac Lab sweep, 4380 episodes per policy (8 gaits $\times$ 5 $\times$ 5 $\times$ 5 command grid $\times$ 5 seeds). Marker = mean of the five seed means, whisker = std over seeds, dashed line = 69-dim baseline. Axes are zero-anchored (ratio scales) or floored well below the data (percentages), so a flat series reads as flat.

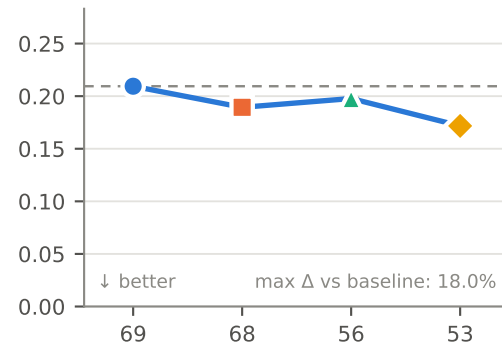
survival rate [%]



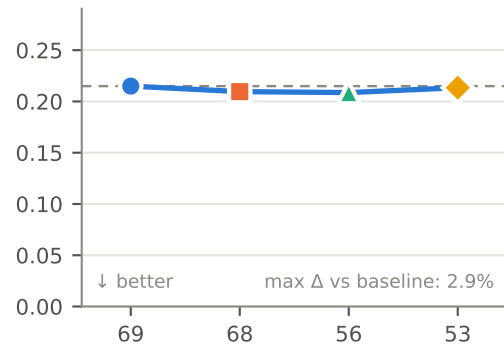
gait contact accuracy [%]



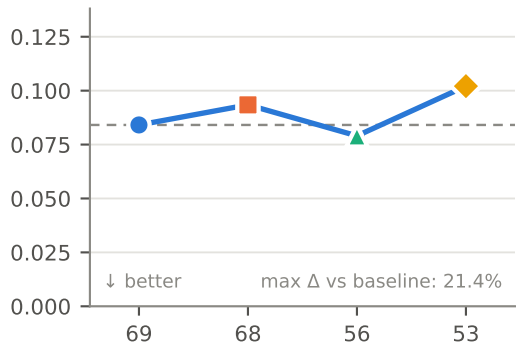
$|e_{v_x}|$ [m/s]



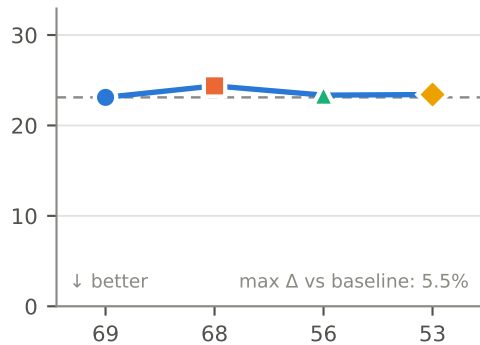
$|e_{v_y}|$ [m/s]



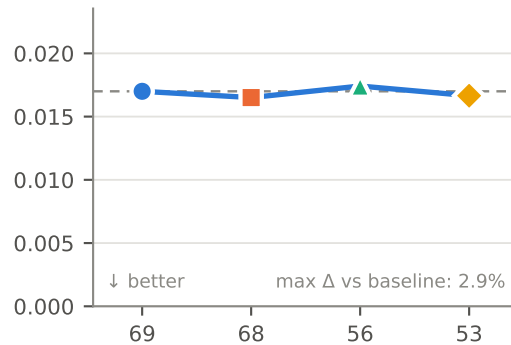
$|e_{\omega_z}|$ [rad/s]



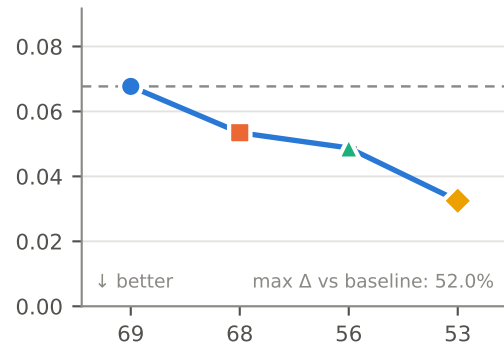
$\|\tau\|$ [Nm]



σ_h [m]



$|\theta|$ [rad]



observation dimension

observation dimension

observation dimension

observation dimension

69: full observation 68: – base height 56: – joint torques 53: – base lin. vel. stand excluded from the averages: it is evaluated at the zero command only.